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High-Resolution Comparative Genomics and Compositional Analysis of WRKY Transcription Factors Across Seven Cereal and Model Plant Genomes

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Abstract

Background

WRKY transcription factors are major regulators of plant stress responses and development, yet their evolutionary dynamics across major cereals and dicot needs further characterization. Previous studies cataloged WRKY genes individually in single species, but no comprehensive comparative analysis integrating phylogenomic, syntenic, and compositional analyses across the monocot-dicots divide has been conducted. This knowledge gap limits our ability to identify conserved functional constraints versus lineage-specific evolutionary innovations in WRKY regulatory networks.

Results

A high-resolution comparative genomic analysis of 547 WRKY genes was performed across seven plant genomes: *Arabidopsis thaliana*, *Oryza sativa* subspecies *japonica* (126 genes), *indica* (109 genes), and the previously uncharacterized *O. glaberrima* (51 genes), *Brachypodium distachyon* (56 genes), *Zea mays (*105 genes), and *Triticum aestivum* (52 genes). Phylogenetic analysis revealed distinct group proportions, with Group III representing 70% of total WRKY genes in cereals compared to only 20.8% Group I and 12.8% Group II, representing a monocot-specific expansion. Group III genes constitute the dominant WRKY classification across all cereal species examined, with pronounced enrichment in cereals (mean 64.9%; range 59.6–69.8%) representing a 2.6-fold difference relative to Arabidopsis (25.0%). This cereal-specific expansion is mechanistically driven by tandem duplication events significantly enriched for Group III genes (Fisher's exact test, $p = 0.007$), maintained under strong purifying selection (mean Ka/Ks = 0.141). Synteny analysis identified 218 collinear gene pairs between rice and *Brachypodium*, 186 with *Zea mays*, 164 with *Triticum aestivum*, and 142 with Arabidopsis, indicating lineage-specific conservation patterns. Evolutionary rate analysis revealed highly conserved WRKY domains (Ka/Ks = 0.08-0.12) juxtaposed against rapidly evolving flanking regions (Ka/Ks = 0.42-0.78), suggesting strong purifying selection on DNA-binding function. t-SNE analysis identified 22 bridge genes with intermediate compositional profiles spanning the monocot-dicot divide, distributed across four cereal lineages and exhibiting structural properties consistent with a directional evolutionary trajectory from ancestral Group I to derived Group III configurations. Notably, *O. glaberrima* showed reduced Group I representation (9.8%) and elevated Group III proportion (80.4%), indicating lineage-specific retention patterns during independent domestication.

Conclusions

This comprehensive analysis establishes a quantitative framework for dissecting WRKY gene family evolution in cereals, identifies stress-responsive orthologs prioritized for crop improvement, and demonstrates that ancient polyploidy, recent segmental duplication, and differential selection pressure collectively shape cereal regulatory architecture. The study provides a foundation for targeted breeding strategies to enhance climate resilience in major cereal and dicot crops.

Keywords WRKY transcription factors, Comparative genomics, Synteny analysis, Gene duplication, Multivariate analysis, *Oryza sativa*, Cereal crops

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Introduction

Transcription factors (TFs) serve as the master architects of plant adaptation, translating developmental cues and environmental stimuli into complex gene expression programs [1]. Among the plant-specific TF lineages, the WRKY superfamily is distinguished by its evolutionary expansion and pivotal role in regulating immunity, abiotic stress tolerance, and hormone signaling [2-4].